WORKPLACE DIVERSITY ANALYSIS

Employee Levels, Compensation, and Pay Equity

Prepared for the Head of HR • July 2026

10,000 employees analyzed

The Headline Finding

There is a real 14% pay gap between men and women. We find no significant gap within the same level and department — the pattern points to representation and promotion differences that show up downstream as a pay gap.

$27,562

Raw pay gap (men earn more)

14.0% higher • statistically significant, p < 0.001

–$703

Adjusted gap, same level & dept

Not significant, p = 0.65 — no significant gap within the same level and department (no performance control)

18.8% vs 31.4%

Share of women vs. men working in Engineering

The highest-paying department, p < 0.001

Data & Method

1

Two data tables

company_hierarchy (employee, boss, department) and employee (salary, degree, sex, experience, signing bonus) — 10,000 employees, no missing or duplicate records.

2

Levels derived from the org chart

Six levels built bottom-up from reporting lines: IC → MM → D → VP → E → CEO.

3

Salary modeled with OLS regression

OLS implemented directly in NumPy; significance from the normal approximation, accurate at n ≈ 10,000.

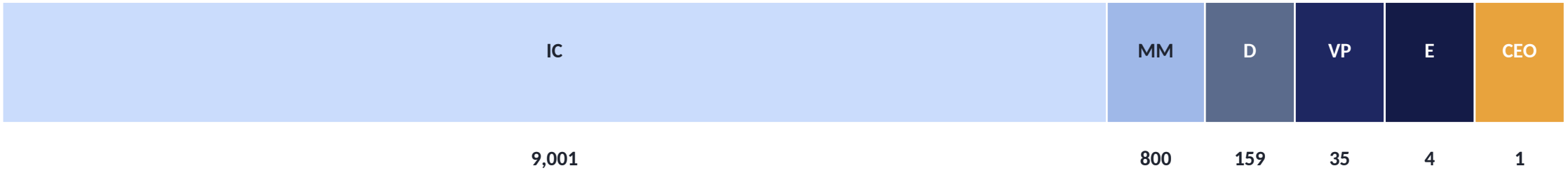
4

Fairness tested, not assumed

Compared raw pay gaps to gaps after controlling for level, department, degree, experience, and signing bonus.

Six Levels, One Pyramid

Share of the 10,000-person company at each level (bar not to scale)



IC = Individual Contributor • MM = Middle Manager • D = Director • VP = Vice President • E = Executive

Levels are defined recursively from the reporting hierarchy: an IC manages no one; each level up manages only people from the level directly below it.

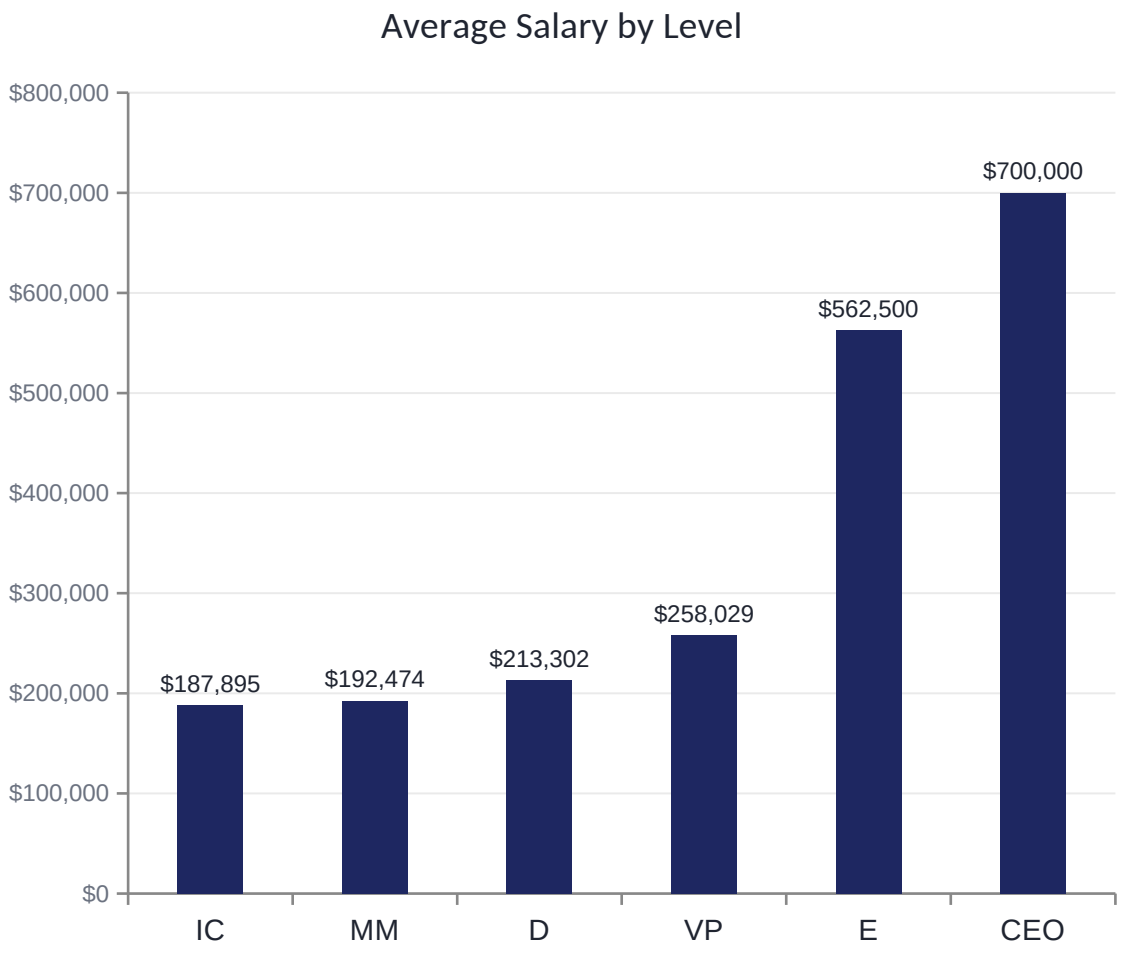
Span of Control (direct + indirect reports)

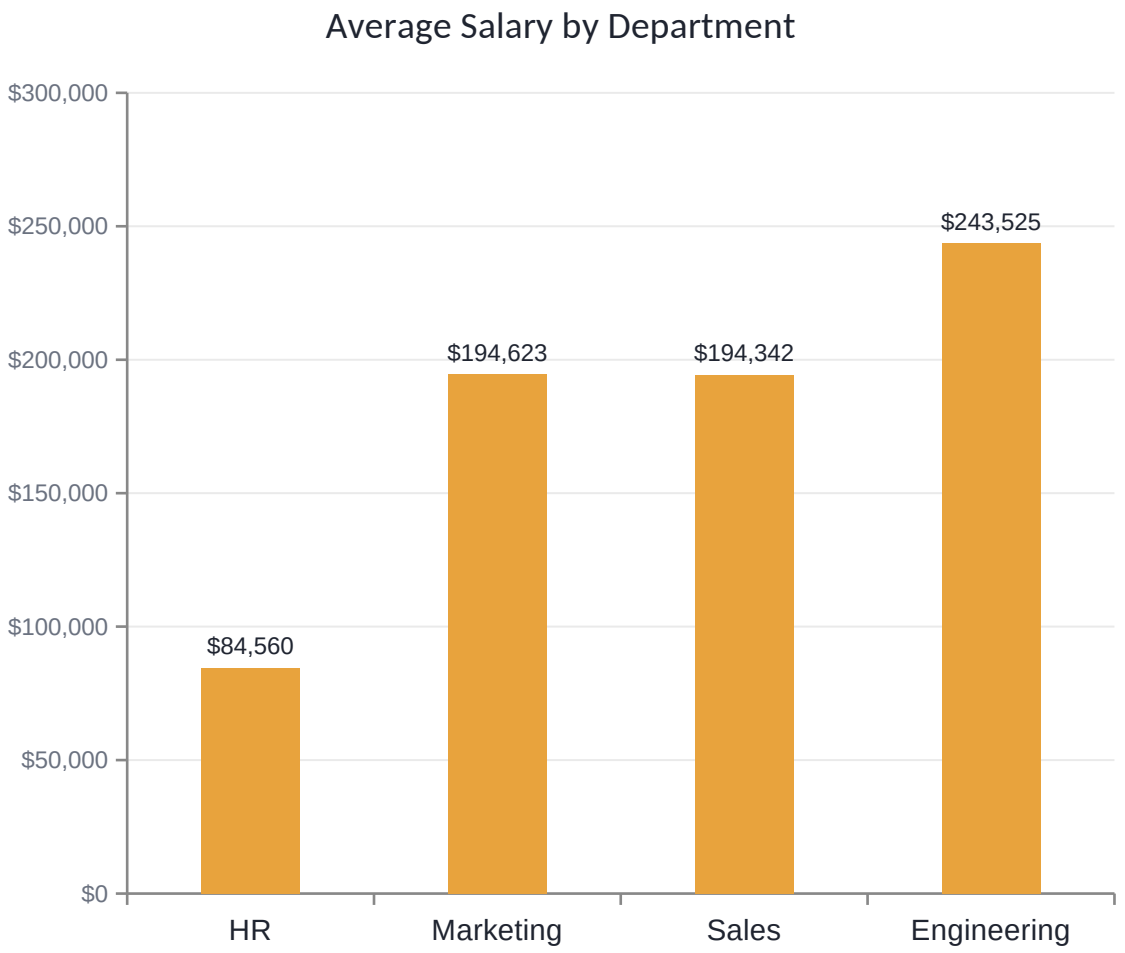
Level	Min	Max	Average
MM	2	23	11.3
D	6	166	61.6
VP	111	607	284.6
E	1,693	3,598	2,498.8
CEO	9,999	9,999	9,999

Sanity check: the CEO manages 9,999 people — everyone else in the company.

What Drives Salary?

Level and department dominate; experience, degree, and signing bonus show no significant effect

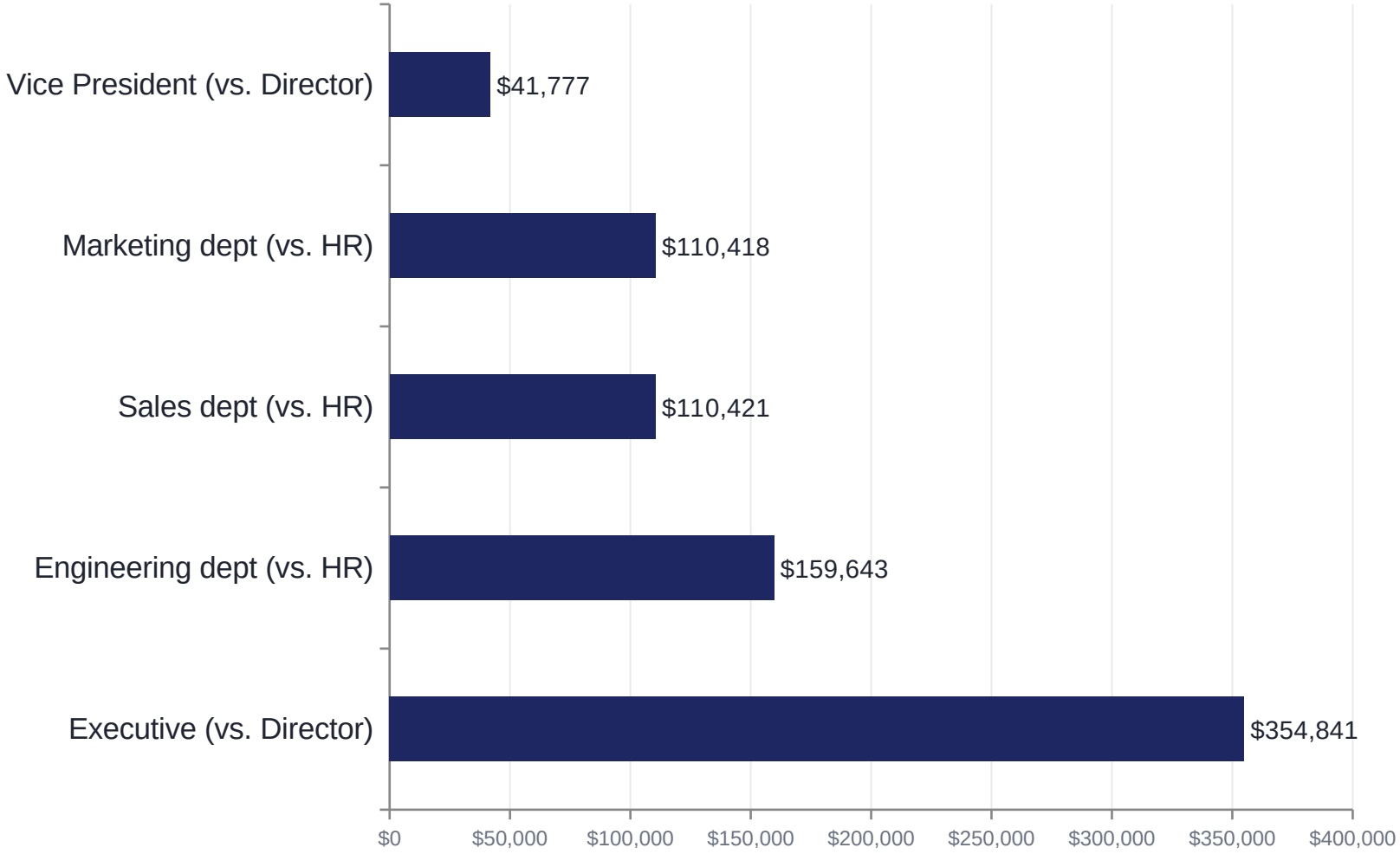




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The Salary Model

Effect of each factor on salary, holding everything else constant ($R^2 = 0.35$, n = 9,999)



Reference employee:

Director, HR dept, High School diploma, Female, 0 yrs experience, no signing bonus

Statistically significant:

Level and department, all $p < 0.01$

Not significant:

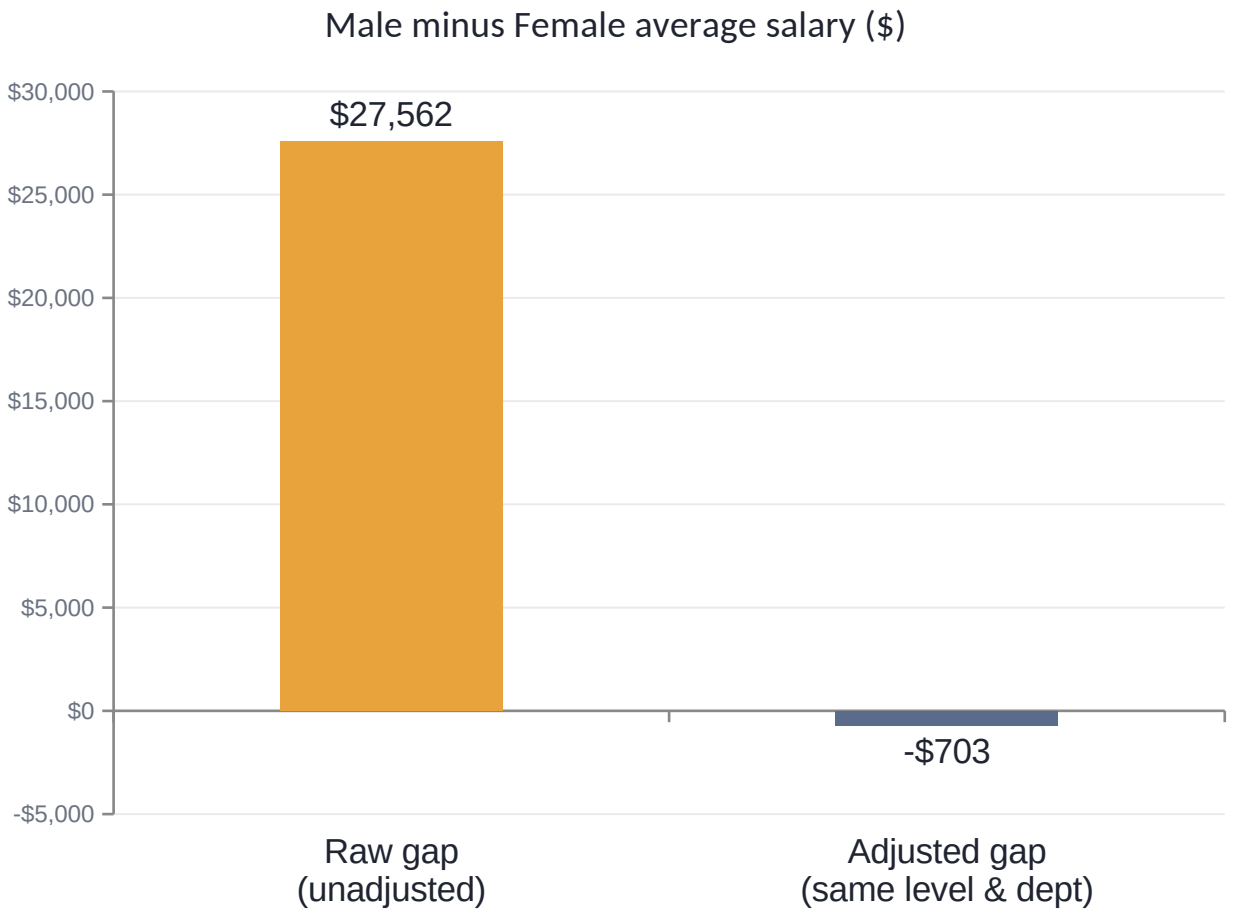
Degree, experience, signing bonus, and sex — all $p \geq 0.13$. Sex in particular: $p = 0.65$.

Bottom line: level and department set pay. With them held constant, sex has no significant effect.

Individual Contributors and Middle Managers earn $16,000–$19,000 less than Directors ($p < 0.01$) — a modest, expected step down the ladder, not shown above to keep the scale readable.

The Pay Gap, Explained

A real raw gap that (almost) fully disappears once level and department are held constant



$27,562 gap

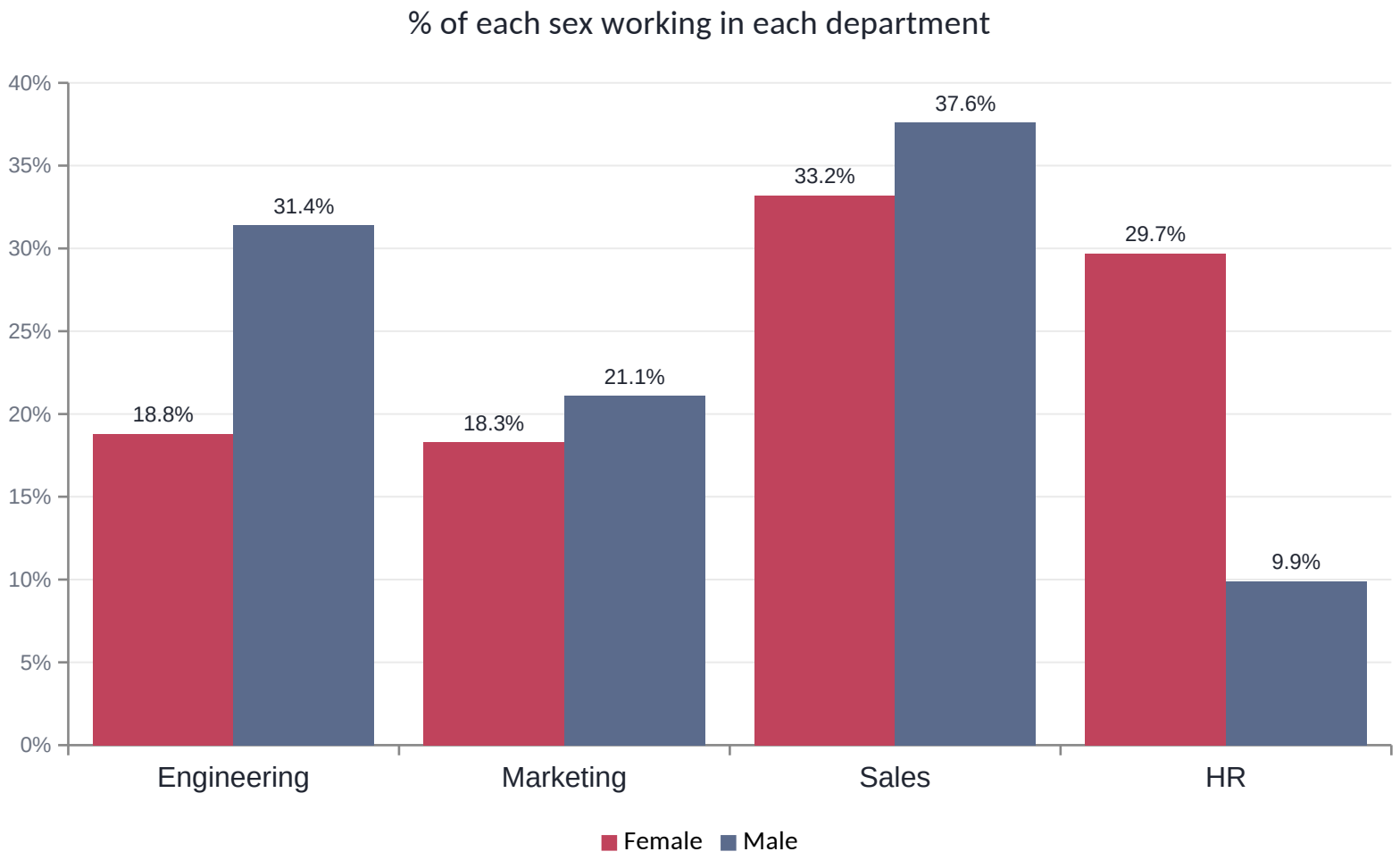
Statistically significant, p < 0.001 — men earn 14.0% more on average, company-wide.

–$703 gap

Not significant, p = 0.65 — no detectable pay difference for the same level and department.

The Real Gap: Representation

A larger share of women than men work in the lowest-paying department, and a smaller share in the highest-paying one



Engineering (highest pay: $243,525 avg)

18.8% of women vs. 31.4% of men

p < 0.001

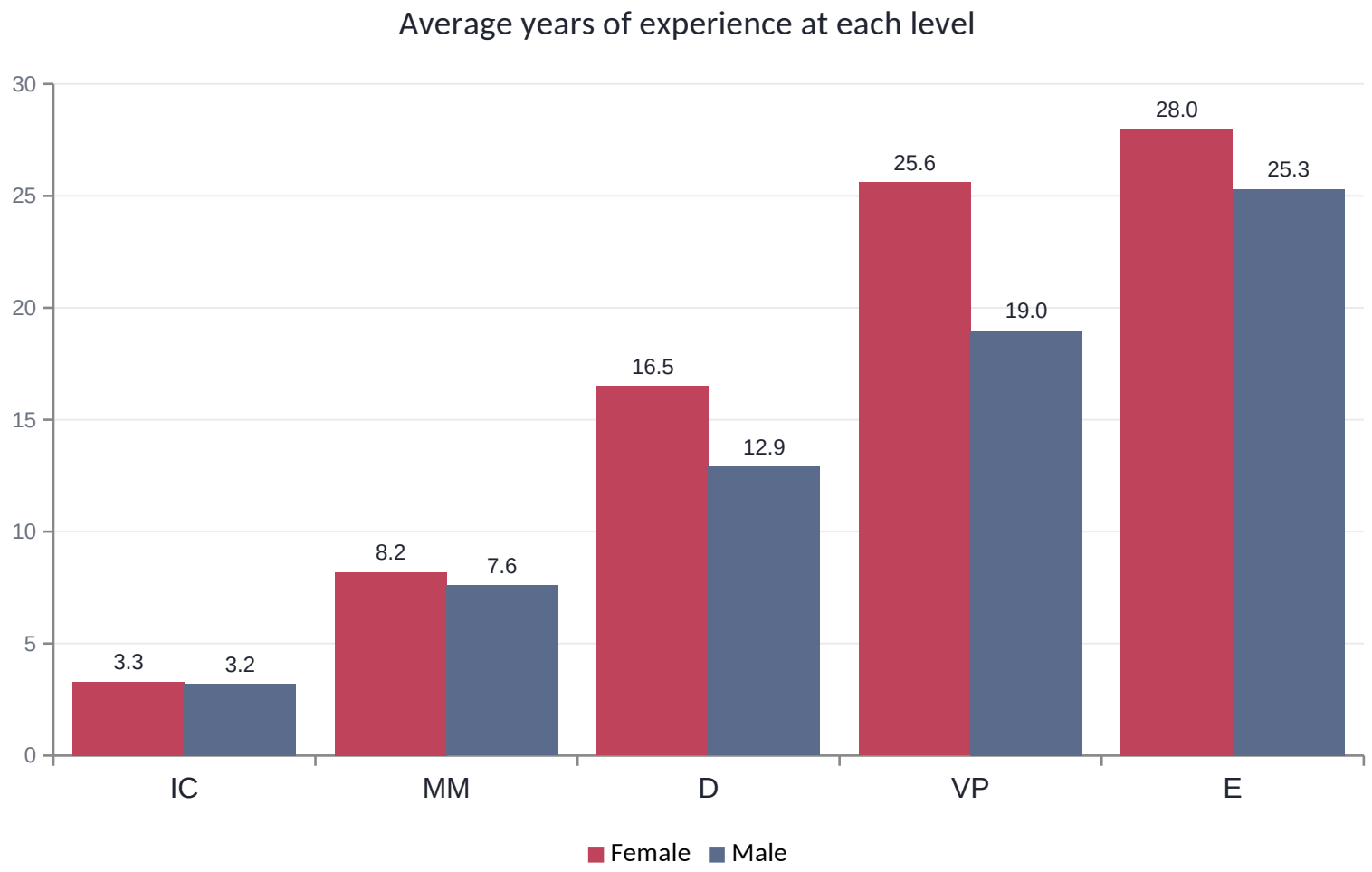
HR (lowest pay: $84,560 avg)

29.7% of women vs. 9.9% of men

p < 0.001

Experience at Senior Levels

Women at senior levels have more experience on average than men at the same level



Director level

16.5 yrs (women) vs. 12.9 yrs (men) (not significance-tested)

Vice President level

25.6 yrs (women) vs. 19.0 yrs (men) (not significance-tested)

1.60% of women reach Director or above, vs. 2.21% of men (p = 0.038).

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Recommendations for the Head of HR

1. **Don't stop at “equal pay for equal role.”**

   The model finds no significant gap within level and department — not proof of fairness (no performance control). It can mask who gets into which department and who gets promoted.

2. **Investigate the Engineering pipeline.**

   Audit hiring, transfer, and retention funnels into the highest-paying department.

3. **Audit promotion timelines and criteria.**

   Review calibration and manager decisions for bias; track time-to-promotion by sex and level.

4. **Set representation targets and monitor over time.**

   Track department mix and leadership representation on a recurring cadence, not a one-time check.

5. **Broaden and repeat the analysis.**

   Extend to other protected characteristics as data allows, and re-run periodically to track progress.

Bottom line

No significant pay gap within level and department. Representation and promotion speed are where to look next.

Questions & discussion welcome.